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25 August 1981
SUPERSEDING
MIL-H-10368C
25 April 1963

MILITARY SPECIFICATION

HAIR BOUND: FOR PADDING ON MILITARY VEHICLES

This specification is approved for use by the Natick Research and Development Laboratories, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bound hair suitably cured in blocks for use as padding on military vehicles.

1.2 Classification. Bound hair shall be of the following classes as specified (see 6.2).

Class 1 - Medium density
Class 2 - Heavy density

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-H-111	-	Hair Curled
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.3).

* 3.2 Material (see 6.4).

3.2.1 Bound hair. The bound hair shall consist of 50 ± 5 percent by weight of animal hair, bound with 50 ± 5 percent by weight of binding elastomer (see 3.2.1.2).

3.2.1.1 Hair. The hair shall be 20 ± 5 percent cattle-tail hair and 80 ± 5 percent winder hog hair as specified in C-H-111 and a length conforming to grade V of C-H-111.

3.2.1.2 Binding elastomer. The binding material shall be polychloroprene (CR) or natural rubber or natural rubber with styrene-butadiene rubber (SBR), in conformance to the requirements of this specification.

3.3 Physical properties.

* 3.3.1 Initial thickness. Unless otherwise specified, the thickness of pads of bound hair, prior to tests involving deformation shall be in multipules of 1 ± 1/8 inch as specified (see 6.2). The thickness shall be measured as specified in 4.5.1.

3.3.2 Density. When determined by the method specified in 4.5.2, the density of the bound hair shall conform to the following ranges:

Class	Grams per cubic centimeter	Pounds per cubic inch
1 - Medium density	0.030 to 0.040	0.0011 to 0.0014
2 - Heavy density	0.039 to 0.052	0.0014 to 0.0019

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* 3.3.3 Compressibility. When determined as specified in 4.5.3 the load required to compress the bound hair to 50 percent of the initial thickness shall be:

Class	Test load-pounds	Pounds per square inch
1	5.5 to 10	0.344 to 0.625
2	13 to 25	0.8125 to 1.562

3.3.4 Set. The reduction in thickness of the bound hair shall be not more than 40 percent of the initial thickness, after being tested as specified in 4.5.4.

* 3.3.5 Form. Unless otherwise specified, the bound hair shall be furnished in slabs approximately 1 by 24 by 72 inches in size.

* 3.3.6 Resistance to low temperature. Bound hair shall not fracture and shall not take more than 25 percent of set when loaded to cause 50 percent reduction in thickness at a temperature of -65°F (see 4.5.5).

3.4 Workmanship. The bound hair shall be uniform in quality, and free of all kinds of foreign matter. It's processing and cure shall be in accordance with good commercial practices.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I, and tested for the characteristics specified in table II. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item inspection.

4.4.2.1 End item examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of slabs. The sample unit shall be one slab. The inspection level shall be I and the acceptable quality level (AQL) shall be 1.5 defects per hundred units.

TABLE I. End item defects

Defect
Wrong size slab
Slab not uniform
Contains foreign matter
Binder not cured properly

4.4.2.2 End item testing. The end item shall be tested for the characteristics listed in table II. The lot size shall be expressed in units of slabs. The sample unit shall be one slab. The inspection level shall be S-1. Any test failure shall be cause for rejection of the lot.

TABLE II. End item tests

Characteristics	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as
Initial thickness	3.3.1	4.5.1	1	Pass or fail
Density	3.3.2	4.5.2	3	Pass or fail
Compressibility	3.3.3	4.5.3	3	Pass or fail
Set	3.3.4	4.5.4	1	Pass or fail
Resistance to low temperature	3.3.6	4.5.5	1	Pass or fail

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* 4.4.3 Packaging inspection. An examination shall be made to determine that packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully packaged except that it need not be closed. Examination for closure defects listed below shall be made on shipping container fully packaged except that it need not be closed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

TABLE III. Packaging defects

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Workmanship	Inadequate application of components, such as: incomplete closure of flaps, loose strapping or taping, or inadequate stapling. Bulged or distorted container.

* 4.5 Methods of inspection. Unless otherwise specified, tests shall be made with stabilized temperature of specimen and ambient air of 75 ± 5°F. The moisture content of the specimen shall be stabilized by storage of the specimen in air having a relative humidity of 50 ± 5 percent for not less than 24 hours. The test specimens shall be pieces 4 inches wide and 4 inches long, cut from the sample selected. The specimens shall be tested in the thickness as received, except that specimens having a nominal thickness of less than 2 inches shall be tested in combination, as required, to provide a minimum test thickness of 2 inches.

4.5.1 Initial thickness. The test specimen shall be placed upon a horizontal plane (flat) surface and, a suitable 5 by 5 inch flat piece of metal, plywood, or plastic shall be centered on top of the specimen. Upon this plate shall be placed weight that will apply to the specimen a total compressive load of 4 pounds (1/4 pound per square inch of specimen area). The initial thickness shall be the average of the height measurements taken at the four corners of the 5 by 5 inch plate, holding the required test load.

4.5.2 Density. The density of the material shall be determined by dividing the weight of a specimen (see 4.5) expressed in grams or pounds, by the volume of the specimen, expressed in cubic centimeters or cubic inches, respectively, as determined by positive measurement of length, width, and initial thickness (see 4.5.1). The average of three such determinations shall be used to determine compliance with 3.3.2.

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4.5.3 Compressibility. The load required to cause 50 percent reduction in the initial thickness (see 4.5.1) of the specimen shall be determined. The method of determining thickness under load shall be as specified in 4.5.1.

4.5.4 Set. The specimen shall be compressed to 50 percent of its initial thickness (see 4.5.1) and retained at this reduced thickness for 48 hours at temperatures of 150 ± 2°F. The compressive load shall then be removed and the specimen allowed to stand 10 ± 1 minutes at room temperature. The set shall be the relationship between the reduction in thickness (after the 10 minute recovery period) and the initial thickness, both measured, as specified in 4.5.1 and expressed as percentage loss in thickness.

4.5.5 Resistance to low temperature. The test specimen shall be measured for initial thickness at laboratory temperature (see 4.5.2), thereafter cooled to -65 ± 2°F, and held at that temperature for 20 hours. It shall then be compressed in a period of approximately 5 seconds, with suitable equipment, to 50 percent of its initial thickness. After release of the compressive load, the recovery set (see 4.5.4) shall be determined at the -65 ± 5°F temperature. This cycle of loading, releasing of the load and measuring of the set shall be repeated two more times. Then the unloaded sample shall be brought to equilibrium at laboratory temperature of 70 ± 5°F, the recovered thickness measured, and the set resulting from loading at low temperature determined (see 4.5.4).

5. PACKAGING

* 5.1 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.1.1 Level A packing. Bound hair shall be packed in a snug-fitting fiberboard shipping container conforming to style optional, grade V2s, or V3s of PPP-B-636, except that dimensional limitations shall be waived. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

* 5.1.2 Level B packing. Bound hair shall be packed in a snug-fitting fiberboard shipping container conforming to style optional, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636, except that dimensional limitations shall be waived. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.3.

* 5.1.2.1 Weather-resistant shipping containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be as specified in 4.4.3.

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* 5.1.3 Commercial packing. Bound hair shall be packed in accordance with MIL-STD-1188.

* 5.2 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The bound hair is intended for use in filling seat cushions, seat backs and pads in military vehicles and for other similar applications. Class 1 is suitable for seat backs, while class 2 is suitable for seat cushions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class of hair to be furnished (see 1.2).
(c) Dimensions of slabs from applicable drawing (see 3.3.1 and 3.3.5).
(d) When a first article sample is required (see 3.1, 4.3 and 6.3).
(e) Selection of applicable levels of packing (see 5.1).
(f) Special marking if required (see 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of bound hair of the same class. The contracting officer should include specific instructions in all procurement instruments regarding arranagements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when praactical as long as it meets the requirements of the specification (see 3.2).

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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